

Technology Management (MG-2007)

Course Instructor(s):

Ms. Zoya Mashadi

Sections CS-8A, CS-8B

Sessional-II Exam

Total Time (Hrs): 01

Total Marks: 40

Total Questions: 04

Roll No

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Course Section

Student Signature

Attempt all the questions.

[CLO 1: Explain the strategic role of technology and innovation management in achieving organizational competitiveness.]

Q1 Write the correct option for MCQs on Answer sheet. No overwriting or correction allowed [10 marks]

1. Strategy is best defined as:

- A. A short-term operational plan
- ☒ B. A long-term plan for achieving success
- C. A daily task allocation system
- D. A financial budgeting tool

2. The transformation of Amazon from an online bookstore to a cloud computing leader is an example of:

- A. Cost leadership
- ☒ B. Strategic diversification
- C. Market failure
- D. Supplier integration

3. Which of the following is NOT one of Porter's Five Forces?

- A. Threat of new entrants
- B. Bargaining power of buyers
- ☒ C. Government regulation
- D. Rivalry among competitors

4. High bargaining power of suppliers generally leads to:

- A. Lower input costs
- B. Increased profitability for firms
- ☒ C. Higher input costs and reduced margins.
- D. No impact on firms

5. IBM is often cited as an example of ambidexterity because it:

- A. Focuses only on hardware production
- ☒ B. Simultaneously exploits existing businesses and explores new technologies
- C. Avoids innovation risks
- D. Operates only in emerging markets

6. Which of the following best represents "exploitation" in organizations?

- A. Experimenting with new business models
- B. Entering unknown markets
- ☒ C. Refining and scaling existing capabilities
- D. Investing in uncertain technologies

7. "Exploration" in strategic management involves:

- A. Improving efficiency of current operations
- B. Minimizing risks
- ☒ C. Experimentation and discovery of new opportunities
- D. Cost reduction

8. Products like Gmail and Google Maps emerged primarily due to:

- A. Strict hierarchical control
- ☒ B. Employee autonomy and slack time
- C. External consultants
- D. Stakeholder involvement

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9. Organizational structure mainly influences:

- A. Employee mindset
- B. Behavior and informal norms
- C. Roles and responsibilities**
- D. Innovation outcomes directly

10. According to Google's Project Aristotle, the most important factor for team effectiveness is:

- A. Technical expertise
- B. Financial incentives
- C. Psychological safety**
- D. Leadership authority

[CLO 2: Analyze the frameworks and processes involved in planning, implementing, reviewing, and controlling innovation initiatives.]

Q2. Case Study - TechNova Solutions

[12 marks]

TechNova Solutions is a mid-sized technology company based in a developing country. It specializes in cloud software, mobile apps, and AI-based business tools for small and medium enterprises (SMEs). Recently, the company has been planning to expand into international markets, especially in Europe and Asia.

However, TechNova is facing several challenges:

- The government has introduced new data protection and cybersecurity laws
- Inflation and currency fluctuations are affecting operational costs
- Customers are demanding more environmentally sustainable technology
- Rapid advancements in AI are increasing competition
- Social media trends are influencing customer expectations
- There are concerns about political instability affecting foreign investment

The CEO has asked the management team to conduct a **PESTEL analysis** to better understand the external environment before expanding.

[CLO 1: Explain the strategic role of technology and innovation management in achieving organizational competitiveness.]

Q3 Short Questions

[6 marks]

- a) Difference between sourcing strategy by making, allying, buying (3 marks)
- b) Difference between resource dependency theory (RDT) vs transaction cost economics (TCE) (3 marks)

[CLO 4: Apply appropriate tools and techniques for developing organizational capabilities in managing technology and fostering innovation.]

[12 marks]

Q4. Explain with examples development process models

- a) Waterfall Model (3)
- b) Critical Path Model (3)
- c) Stage Gate Model (3)
- d) Agile/Hybrid Models (3)

Below is the **complete solved paper** based on the attached exam paper.

Q1: MCQs — Correct Options

1. **B — A long-term plan for achieving success**
 2. **B — Strategic diversification**
 3. **C — Government regulation**
 4. **C — Higher input costs and reduced margins**
 5. **B — Simultaneously exploits existing businesses and explores new technologies**
 6. **C — Refining and scaling existing capabilities**
 7. **C — Experimentation and discovery of new opportunities**
 8. **B — Employee autonomy and slack time**
 9. **C — Roles and responsibilities**
 10. **C — Psychological safety**
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Q2: Case Study — PESTEL Analysis of TechNova Solutions

Political:

TechNova faces political instability, which can affect foreign investment, international expansion, business confidence, and long-term planning. Unstable policies may create uncertainty for expansion into Europe and Asia.

Economic:

Inflation and currency fluctuations are increasing operational costs. This can reduce profit margins, make pricing difficult, and affect TechNova's ability to compete internationally.

Social:

Social media trends are influencing customer expectations. Customers now expect faster service, better user experience, sustainability, and transparency from technology companies.

Technological:

Rapid technological advancements are increasing competition. TechNova must continuously innovate in cloud software, mobile apps, and AI-based solutions to remain competitive.

Environmental:

Customers are demanding more environmentally sustainable products and services.

TechNova should adopt green technology practices, energy-efficient cloud solutions, and sustainable business operations.

Legal:

New data protection and cybersecurity laws create compliance challenges. Since TechNova deals with cloud software and AI business tools, it must ensure data privacy, cybersecurity compliance, and proper legal controls before entering new markets.

Conclusion:

Before expansion, TechNova should strengthen compliance, manage currency and cost risks, invest in innovation, monitor political stability, and align its products with sustainability and customer expectations.

Q3: Short Questions

a) Difference between Sourcing Strategy by Making, Allying, and Buying

Making means developing technology, products, or capabilities internally using the company's own resources. It gives greater control and protects knowledge, but it is costly and time-consuming.

Allying means collaborating with another firm through partnerships, joint ventures, or strategic alliances. It helps share risk, access external expertise, and enter markets faster.

Buying means acquiring technology, resources, or complete firms from outside. It is the fastest option but may be expensive and can create integration challenges.

b) Difference between Resource Dependency Theory and Transaction Cost Economics

Resource Dependency Theory — RDT:

RDT focuses on how firms depend on external resources such as suppliers, partners, technology providers, investors, and regulators. Firms use alliances, mergers, or partnerships to reduce uncertainty and gain control over critical resources.

Transaction Cost Economics — TCE:

TCE focuses on the cost of making exchanges in the market, such as negotiation, monitoring, contracts, and enforcement. Firms decide whether to make internally, buy from the market, or form alliances based on which option reduces transaction costs.

Main Difference:

RDT is about **managing dependence and power over resources**, while TCE is about **minimizing the cost and risk of transactions**.

Q4: Development Process Models with Examples**a) Waterfall Model**

The Waterfall Model is a linear and sequential development process. Each stage must be completed before moving to the next stage. The main stages include requirements, design, development, testing, implementation, and maintenance.

Example:

A government payroll system can use the Waterfall Model because requirements are usually fixed and clearly documented before development starts.

Strength:

It provides clear documentation, structure, and control.

Weakness:

It is not flexible when requirements change during the project.

b) Critical Path Model

The Critical Path Model identifies the longest sequence of dependent activities in a project. These activities determine the minimum time required to complete the project. Any delay in the critical path delays the entire project.

Example:

In launching a new software product, activities such as requirement finalization, coding, testing, and deployment may form the critical path. If testing is delayed, the product launch will also be delayed.

Strength:

It helps managers identify important tasks and control project timelines.

Weakness:

It may not handle uncertainty or changing project conditions very well.

c) Stage-Gate Model

The Stage-Gate Model divides innovation or product development into different stages. At the end of each stage, there is a “gate” where management reviews progress and decides whether to continue, improve, pause, or stop the project.

Example:

A company developing a new mobile application may follow stages such as idea generation, concept development, prototype, testing, launch, and review. At each gate, management checks feasibility, cost, risk, and market potential.

Strength:

It reduces risk by reviewing projects step by step.

Weakness:

It can slow down innovation if approvals take too much time.

d) Agile / Hybrid Models

Agile is an iterative and flexible development approach. Work is divided into short cycles called sprints. Teams continuously test, improve, and adapt based on feedback.

Example:

A startup developing an AI chatbot may use Agile because user requirements can change frequently, and continuous feedback is needed.

Hybrid Model:

A Hybrid Model combines traditional planning with Agile flexibility. For example, a company may use Waterfall for budgeting and documentation, while using Agile for software development.

Strength:

It allows flexibility, faster feedback, and continuous improvement.

Weakness:

It requires strong communication, team coordination, and active customer involvement.